

Empirical Analysis of the Business, Economic, or Societal Impact of Artificial Intelligence (AI)

1. Project Overview

The final project is an **individual, end-to-end data analytics study** conducted using Python. It aims to develop technical proficiency in Python while introducing students to the structure and discipline of academic research.

Students will complete the full research process: formulating a meaningful research question, developing testable hypotheses, designing a feasible data strategy, collecting and preparing data, conducting systematic analysis, interpreting results, and communicating findings in a structured format. This sequence reflects how empirical research and professional analytical projects are carried out in practice.

This project also strengthens practical Python and data analysis skills. Techniques in coding, data cleaning, visualization, and modeling developed here are transferable to future coursework, independent research, internships, and graduate study. The final output may serve as a foundation for a senior thesis, conference submissions, or more advanced empirical work.

The project progresses **cumulatively** and **iteratively** through three milestone assignments, ensuring steady development, methodological rigor, and reduced last-minute risk. It concludes with **a written term paper** and **an in-class presentation**.

2. Milestone Assignments

The three milestone assignments guide students through the early stages of academic research and empirical project development. **Each stage builds on the previous one** and prepares students for the final term paper.

2.1. Assignment 1 – Research Question Formulation

Deadline: 15 March (Sunday), 23:59, Beijing Time

Deliverables: A concise proposal (3–4 pages) (12-font, 1.5-spaced) including:

- Clearly defined and narrowly scoped research question
- Motivation and research gap
- Short literature review
- Preliminary thoughts on possible data sources
- Reference

Purpose: To initiate the research process by helping students narrow a broad area of interest into a focused and researchable academic question.

Focus: Students should select a small, well-defined topic related to the business, economic, or societal impact of AI. The topic must be specific rather than broad (e.g., not “AI and business,” but a clearly bounded issue within that domain).

Students are expected to:

- Read several related academic papers (reading abstracts is sufficient at this stage) to understand how research questions are framed.
- Use AI tools, if helpful, to gain an initial overview of the topic and identify potential research directions.
- Learn what constitutes an academic research question.
- Propose their own research question.
- Clearly articulate the motivation and identify a potential research gap.
- Provide a short and focused literature review of the selected sub-topic.

Students are encouraged to draw inspiration from studies examining AI’s impact **using digital trace data from online communities and platforms** (e.g., GitHub, Stack Overflow, freelancer platforms, etc.).

Recommended Readings:

- Bertomeu, J., Lin, Y., Liu, Y. and Ni, Z., 2025. [The impact of generative AI on information processing: Evidence from the ban of ChatGPT in Italy](#). *Journal of Accounting and Economics*, p.101782. [[Working Paper Version](#)]
- del Rio-Chanona, R.M., Laurentsyeve, N. and Wachs, J., 2024. [Large language models reduce public knowledge sharing on online Q&A platforms](#). *PNAS Nexus*, 3(9), p.400.
- Demirci, O., Hannane, J., and Zhu, X., 2025. [Who is AI replacing? The impact of generative AI on online freelancing platforms](#). *Management Science*, 71(10), pp.8097-8108. [[Working Paper Version](#)]
- Kim, S., Jin, G.Z. and Lee, E., 2026. [Does generative AI crowd out human creators? Evidence from Pixiv](#). *Working Paper*.
- Shan, G. and Qiu, L., 2025. [Examining the impact of generative AI on users’ voluntary knowledge contribution: Evidence from a natural experiment on stack overflow](#). *Information Systems Research*. 0(0). [[Working Paper Version](#)]
- Song, F., Agarwal, A. and Wen, W., 2024. [The impact of generative AI on collaborative open-source software development: Evidence from GitHub Copilot](#). *Working Paper*.

Useful Resources

- A tool that may help you identify related literature: [Connected Papers](#)
- Working paper websites: [SSRN](#), [NBER](#), [arXiv](#)

- [UTD 24 journal list](#)

2.2. Assignment 2 – Hypotheses, Data and Empirical Strategy

Deadline: 12 April (Sunday), 23:59, Beijing Time

Deliverables: A refined proposal including: (8-10 pages, 12pts, 1.5-spaced, **please use the attached template**)

- Introduction
- Literature Review
- Hypotheses Development
- Data Sources and Collection Plan
- Empirical Strategy Plan
- References

Purpose: Assignment 1 revealed several common issues, including overly broad research questions, weak connections to the literature, and unclear feasibility. Assignment 2 is designed to address these limitations by strengthening specificity, testability, and execution readiness. This assignment moves your project from a **research idea** to a **research design**. You are expected to:

- Refine and sharpen your research question into a specific, testable problem
- Rebuild your literature review into a coherent analytical narrative
- Translate your idea into clear hypotheses
- Define constructs and measurements
- Identify real, feasible data sources
- Design a basic empirical strategy (e.g., what kind of analysis)

2.3. Assignment 3 – Data Collection Report & Research Readiness

Deadline: 10 May (Sunday), 23:59, Beijing Time

Purposes: Assignment 2 moved your project from a research idea to a research design. Assignment 3 moves it from design to execution. By the time you submit this assignment, you should have **actual data in hand and be ready to begin analysis**.

This assignment also introduces you to GitHub as a professional tool for managing your research project — version control, documentation, and reproducibility.

Part 1: GitHub Repository (required)

Create a **public** GitHub repository for your research project. The repository should contain the following:

Repository structure:

```

your-repo-name/
├── README.md           ← Project overview (required)
├── data/
│   └── your_dataset.csv ← Your collected dataset (required)
├── code/
│   └── data_collection.ipynb ← Your data collection notebook (required)
└── docs/
    └── data_description.md ← Detailed data description (required)

```

1. README.md should include:

- Project title and your name
- Research question (one sentence)
- Brief description of the dataset: what it contains, where it comes from, how it was collected
- Instructions for running the data collection notebook (what packages to install, what credentials are needed if any)

2. data/ folder:

- Your collected dataset in **CSV format**
- If your dataset is too large for GitHub (e.g., >100MB), provide a sample (first 1,000 rows) and note the full dataset size in the README

3. code/ folder:

- The **Jupyter notebook(s)** used to collect your data
- The notebook should be runnable — someone following your README instructions should be able to reproduce the data collection
- Include comments explaining what each major code section does

4. docs/data_description.md should include:

- Data source: which website or API, and the URL
- Collection method: API, HTML scraping, or both
- Collection date and time period covered
- Number of observations (rows) and variables (columns)
- Variable descriptions: for each column, provide the variable name, data type, description, and an example value
- Any known data quality issues (missing values, duplicates, encoding issues)

Part 2: Canvas Submission (required)

Submit a **short report** (3–4 pages, 12pt font, 1.5-spaced) on Canvas that includes:

1. **Research question** (restated from Assignment 2, refined if needed)
2. **Data collection summary**

- What data did you collect? From which source?
- Which method did you use — API, HTML scraping, or both? Why?
- How many observations and variables?
- What challenges did you encounter and how did you solve them?

3. Dataset overview

- A summary table of your variables (name, type, description)
- Basic descriptive statistics (mean, min, max, count) for key numeric variables
- Note any data quality issues that will need cleaning

4. Research readiness assessment

- How does your collected data map to your research question and hypotheses from Assignment 2?
- What analysis steps are next? (This previews your work for the data analysis sessions)
- Are there any gaps between what you planned to collect (Assignment 2) and what you actually collected?

5. GitHub repository link

- Provide the full URL to your GitHub repository

3. Final Deliverables

The final project concludes with two deliverables: a written term paper (15%) and an in-class presentation (15%), together accounting for 30% of the course grade. Both deliverables build directly on the work completed in Assignments 1–3 and should reflect iterative refinement based on instructor feedback.

3.1 Term Paper

Deadline: 10 June (Wednesday), 23:59, Beijing Time

Deliverables: A conference-style empirical research paper (15–20 pages including everything, 12-point font, 1.5-spaced, single column) submitted as a Word file or PDF file if you use LaTeX on Canvas.

Purpose: The term paper is the culmination of your semester-long research project. It integrates and extends the work completed across Assignments 1–3 into a cohesive academic paper. You should refine your earlier submissions based on feedback received, incorporate the data analysis techniques covered in Weeks 11–14, and present your findings in a clear, well-structured format.

Paper Structure: Your paper should follow the standard structure of an empirical research paper. Include all of the following sections:

- 1) **Title Page** — Include the paper title, your name, course number (MGS 3001 WHS01), date, **GitHub repository URL (update the repository from Assignment 3 with your notebooks for visualization, and statistical modeling)**, and **abstract**.
- 2) **Introduction** — State the research question, explain its significance and practical relevance, and provide a brief overview of the paper's structure. This section should motivate the reader and clearly articulate the research gap your study addresses (Also refer to the template provided in Assignment 2).
- 3) **Literature Review** — Synthesize relevant academic literature to build a coherent analytical narrative. Organize by themes or theoretical perspectives rather than summarizing individual papers. Clearly position your study relative to existing work.
- 4) **Hypotheses** — Present your hypotheses with clear theoretical justification. Each hypothesis should be specific, testable, and logically derived from the literature review.
- 5) **Data and Methodology** — Describe your data source(s), collection method (API, HTML scraping, or both), sample characteristics (number of observations, time period, key variables), and any data cleaning or transformation steps. Explain your empirical strategy, including the analytical techniques used (e.g., descriptive statistics, visualization, regression analysis) and how they address your hypotheses. This section should be detailed enough for another student to understand and replicate your approach. (**Tables should be used to organize and present data where applicable.**)

- 6) **Results** — Present your findings systematically. Include relevant tables, charts, and/or regression output. Organize results by hypothesis. Use visualizations to support key patterns. Interpret all tables and figures in the text; do not leave them to speak for themselves.
- 7) **Discussion and Implications** — Discuss what your results mean in the context of your research question and existing literature. Address whether your hypotheses were supported, partially supported, or not supported. Discuss practical or managerial implications: what can businesses, policymakers, or practitioners learn from your findings? Acknowledge limitations of your study (e.g., data constraints, methodological choices) and suggest directions for future research.
- 8) **Conclusion** — Provide a brief summary of the study's purpose, key findings, and contributions. This section should be concise (typically one to two paragraphs).
- 9) **References** — List all sources cited in the paper using a consistent citation. Include at least 5 academic references (journal articles, conference papers, or working papers from reputable sources such as SSRN, NBER, or arXiv).

Formatting Requirements (A guideline, not a strict requirement):

- Length: 15–20 pages total (including everything)
- Font: 12-point, Times New Roman or similar serif font
- Spacing: 1.5-spaced
- Margins: 1 inch on all sides
- Single column
- All figures and tables must be numbered, titled, and referenced in the text
- Submit as a Word file or a PDF file if you use LaTeX on Canvas
- Ensure your GitHub repository (from Assignment 3) is up to date and with all final code (separated notebooks for data collection, cleaning, variable construction, visualization, and statistical modeling), data, and documentation. The repository should contain code and data files only — the final paper and figures need not be uploaded.

3.2 Final Presentation

Date: 4 June (Thu) & 9 June (Tue)

Deliverables: An in-class oral presentation (6+2 minutes per project) accompanied by a slide deck (submitted on Canvas before 9:30AM on your presentation day).

Purpose: The final presentation gives you the opportunity to communicate your research findings to your peers in a clear, concise, and professional manner. Presenting research is a core skill in both academic and professional settings. This exercise develops your ability to distill a complex project into key messages, support claims with evidence, and respond to questions.

Format:

- Presentation length: approximately **6 minutes** per student (**scripts are allowed**)
- Q&A: approximately **2 minutes** following each presentation
- Presentations will be held during the scheduled class sessions on 4 June (9 presenters) and 9 June (7 presenters), 10:00 – 11:15 AM in CBPM A102.
- All presentation files will be projected from the instructor's computer; upload your slides on Canvas before 9:30 AM on your presentation day.
- Attendance is mandatory for both presentation sessions; absence will result in a participation deduction

Slide Deck Structure (suggested, approximately 6-10 slides):

- 1) **Title Slide** — Project title, your name, course, and date.
- 2) **Motivation and Research Question** (1 slide) — Why does this topic matter? What specific question are you investigating?
- 3) **Key Literature and Hypotheses** (1–2 slides) — Briefly summarize the **most relevant** prior work and state your hypotheses.
- 4) **Data and Methodology** (1–2 slides) — Describe your data source, collection method, sample size, and key variables. Explain your analytical approach.
- 5) **Results** (1–2 slides) — Present your main findings using charts, tables, or regression output. Focus on **the most important results rather than showing everything**.
- 6) **Discussion and Implications** (1 slide) — What do the results mean? What are the practical or managerial takeaways?
- 7) **Limitations and Future Work** (1 slide) — Acknowledge data or methodological limitations. Suggest what future research could explore.

Presentation Tips:

- Use slides as visual aids, not scripts. Avoid reading directly from your slides.
- Keep text minimal on each slide; use visuals (charts, diagrams) to convey data.
- Practice your timing. A 6-minute presentation requires careful selection of what to include and what to leave out.
- Be prepared to answer questions about your data, methods, and interpretation.

4. Academic Integrity & AI Use

Academic integrity is fundamental to this course and to the research process. All submitted work must represent your own intellectual effort. The following policies apply to all milestone assignments, the term paper, and the final presentation.

General Standards:

- This is an individual project. All research, coding, analysis, writing, and presentation work must be your own.
- All sources — including academic papers, websites, datasets, and any AI-generated content — must be properly cited.
- Plagiarism, fabrication of data or results, and submission of another person's work as your own are serious violations and will be handled in accordance with University academic integrity policies.

AI Use Policy:

The use of AI tools (e.g., ChatGPT, Claude, GitHub Copilot) is permitted and encouraged as a learning aid in this course, consistent with the principle of responsible AI adoption discussed in class. However, AI use must be transparent, purposeful, and subordinate to your own critical thinking. The following guidelines apply:

- 1) **Permitted Uses** — You may use AI tools to assist with brainstorming research ideas, debugging code, improving code efficiency, learning new Python techniques, generating initial drafts for revision, proofreading and grammar checking, creating GitHub repository documentation, and designing presentation slides.
- 2) **Prohibited Uses** — You may not submit AI-generated text as your own writing without substantial revision and intellectual engagement. You may not use AI to fabricate data, generate fake references, or produce analysis that you do not understand. You may not use AI to complete entire sections of the term paper without meaningful personal contribution.
- 3) **Disclosure Requirement** — If you use AI tools in a substantive way (beyond basic spell-checking), you must include a brief disclosure at the end of your submission. State which tool(s) you used, for which tasks, and how you verified or revised the AI-generated output. A simple one-paragraph statement is sufficient. For example: "I used ChatGPT to help debug my web scraping code and to generate an initial outline for the literature review, which I substantially revised and rewrote."
- 4) **The 4D Fluency Standard** — When using AI, apply the 4D framework discussed in class: Delegation (decide what to assign to AI vs. yourself), Description (write clear prompts), Discernment (critically evaluate AI output), and Diligence (use AI responsibly and transparently).

Guiding Principle: The purpose of this project is for you to learn. AI tools should complement your learning process, not replace it. You are expected to understand every part of your submitted work — your code, your analysis, and your writing. During the final presentation, you may be asked to explain any aspect of your project in detail. If you cannot explain your own work, it will be treated as a potential academic integrity concern.